

Chapter

1

Internet, World Wide Web and Applications

- Keeping up-to-date with IT
- Content Management System
- Web Basics
- Introduction to Web 2.0, 3.0 and Future Web

Internet

- The Internet is a **globally connected network system** that facilitates communication and data services through a vast collection of private, public, business, academic, and government networks.
- The large system of connected computers around the world that allows people to share information and communicate with each other.
- It's a great way of providing and accessing information and is available almost all over the world.

WWW

- The World Wide Web, often just called the Web, is a system of interlinked documents and resources, accessed via the Internet.
- Components of WWW:
 - **HTML** (Hypertext Markup Language): The standard markup language for creating web pages.
 - **HTTP/HTTPS** (Hypertext Transfer Protocol/Secure): The protocols for transmitting web pages over the Internet.
 - **URLs** (Uniform Resource Locators): The addresses used to access web resources.
 - **Web Browsers**: Software applications like Chrome, Firefox, and Safari that retrieve, present, and traverse information on the Web.
 - **Web Servers**: Computers that store web pages and serve them to users upon request.



The Web was invented by Tim Berners-Lee in 1989 while working at CERN. His goal was to create a system for sharing information easily among scientists

Web Application

- Web Application (WA) is a software program that runs on a web server and is accessed through a web browser.
- Characteristics of web application:
 - WA can be accessed from any device with an internet connection and a web browser, making them highly versatile and platform-independent.
 - Unlike traditional software, WA do not need to be downloaded or installed on a user's device. They run directly within the browser.
 - Updates and maintenance are performed on the server side, meaning users always access the latest version without manually updating anything.

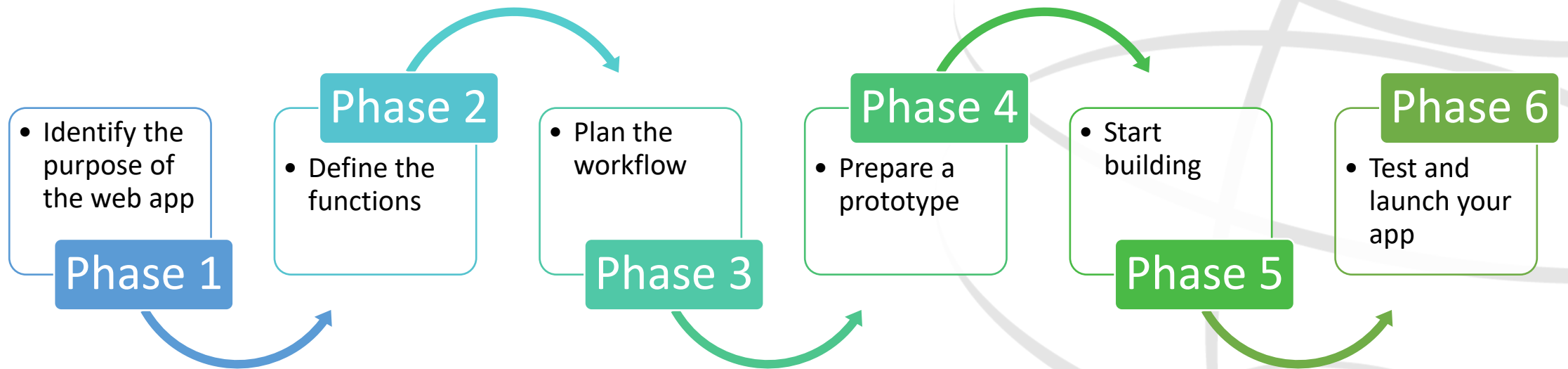
Advantages of Web Application

- Faster and affordable to build compared to native mobile apps
- Easy to maintain and update
- Works on any device with a compatible browser
- Unlike a mobile app, web applications do not have to be installed. They can be used directly from the web browser.
- Web applications are typically more secure since they're built on dedicated servers with a more robust security infrastructure than locally installed apps on individual devices.

Main Types of Web Application



Web Application Development Phases



Technology Advancement

- Progressive Web Application (PWA)
- Blockchain
- Artificial Intelligence
- Generative AI / AI Chatbot
- Serverless architecture
- Improve native cybersecurity
- Optimized integrated voice search
- Machine learning
- Mobile-first design
- Cloud computing
- API-first approach
- Internet of Thing (IoT)

How Does the Future of Web App Look Like

- More dynamic
- Have a layer of interactions that can be further expanded with advanced frameworks and tools
- Augmented reality and artificial intelligence
- Web apps allow businesses to build apps for their different processes on a cloud server
- More adoption of Progressive Web App (PWA)

Content Management System (CMS)

- A CMS is a **software application** that allows users to **build and manage a website without having to code it from scratch** or know how to code at all. With a CMS, you can create, manage, modify, and publish content in a user-friendly interface.
- The main benefit of a content management system is that it allows non-technical people to publish content. This dramatically cuts the cost of maintaining a website.



Advantages & Disadvantages of CMS

Advantages	Disadvantages
• User-friendliness. • Quick deployment and customize • Ease of use and maintenance, including updates. • Cost-efficiency, especially with out-the-box solutions, open source or freeware. • Extendable functionality, through a large number of plugins and extensions. • Search Engine Optimization (SEO) friendly features. • Developer and community support. • Content organization • Multilingual Capabilities • Versatility • Collaboration • Security Policies • Less Expensive	• CMS can have limitations on design and customization. • CMS can have performance issues, especially as a website grows in size and complexity. • CMS platforms can be vulnerable to security concerns. • CMS can have ongoing maintenance and update costs. • Learning curve: Some staff may need web training to understand the processes. • CMS can depend on third-party plugins, which can have update or capability issues. • Open source CMS can be complex and challenging to use and may require technical knowledge or hiring a developer. They can also have compatibility issues with themes, plugins, or hosting services.

Web Application VS Website

Parameter	Web Application	Website
Definition	A website is a group of globally accessible, interlinked web pages which have a single domain name. It can be developed and maintained by an individual, business or organization. The website aims to serve a variety of purposes.	A web application is a software or program which is accessible using any web browser. Its frontend is usually created using languages like HTML, CSS, Javascript, which are supported by major browsers. While the backend could use any programming stack like LAMP, MEAN, etc. Unlike mobile apps, there is no specific SDK for developing web applications.
Created for	A web application is designed for interaction with the end user	A website mostly consists of static content. It is publicly accessible to all the visitors.
User interaction	In a web application, the user not only reads the page content but also manipulates the restricted data.	A website provides visual & text content which user can view and read, but not affect it 's functioning.
Authentication	Web applications need authentication, as they offer a much broader scope of options than websites.	Authentication is not obligatory for informational websites. The user may ask to register to get a regular update or to access additional options. This features not available for the unregistered website visitors.
Task and Complexity	Web application functions are quite higher and complex compared to a website.	The website displays the collected data and information on a specific page.
Type of software	The web application development is part of the website. It is itself not a complete website.	The website is a complete product, which you access with the help of your browser.
Compilation	The site must be precompiled before deployment	The site doesn't need to be pre-compiled
Deployment	All changes require the entire project to be re-compiled and deployed.	Small changes never require a full re-compilation and deployment. You just need to update the HTML code.

Web 1, 2 and 3

No.	Web 1.0	Web 2.0	Web 3.0
1.	Mostly Read-Only	Wildly Read-Write	Portable and Personal
2.	Company Focus	Community Focus	Individual Focus
3.	Home Pages	Blogs / Wikis	Live-streams / Waves
4.	Owning Content	Sharing Content	Consolidating Content
5.	WebForms	Web Applications	Smart Applications
6.	Directories	Tagging	User behavior
7.	Page Views	Cost Per Click	User Engagement
8.	Banner Advertising	Interactive Advertising	Behavioral Advertising
9.	Britannica Online	Wikipedia	The Semantic Web
10.	HTML/Portals	XML / RSS	RDF / RDFS / OWL
11.	Data was not Focused.	Data of many was controlled by some mediatory.	Data was personalized and no use of mediatory.
12.	Information sharing is the goal.	Interaction is the goal.	Immersion is the goal.
13.	It connects information as its primary goal.	It aims to connect people.	Focuses on relating knowledge.
14.	Static websites	Introduction of web applications	Intelligent web-based functions and apps
15.	A simpler, more passive web.	An enhanced social Web	A semantic web exists.
16.	Web and File Servers, HTML, and Portals are technologies connected to Web 1.0.	AJAX, JavaScript, CSS, and HTML5 are examples of related technology.	Web 3.0 technologies include blockchain, artificial intelligence, and decentralized protocols.
17.	Associated Technologies •Web and File Servers •Search Engines (including AltaVista and Yahoo!) •E-mail accounts (Yahoo!, Hotmail) •Peer-to-Peer File Sharing (Napster, BitTorrent) and others.	Associated Technologies •Frameworks for Ajax and JavaScript •Microsoft.NET •Blogs •Wikis and others.	Associated Technologies •Searching Using Semantics •Databases of Information •Ontologies •Intelligent Digital Personal Assistants and others.

Future Web?

- Increasing focus on **User Experience (UX)** and responsive design. Use **Artificial Intelligence (AI)** and **Machine Learning (ML)** in website development. Increasing use of **voice search** and **Natural Language Processing (NLP)**.
- **Progressive Web Apps (PWAs)** are web applications that provide users with an app-like experience on their mobile devices without the need to install or download the application.
- Offline accessibility.
- Push notification.

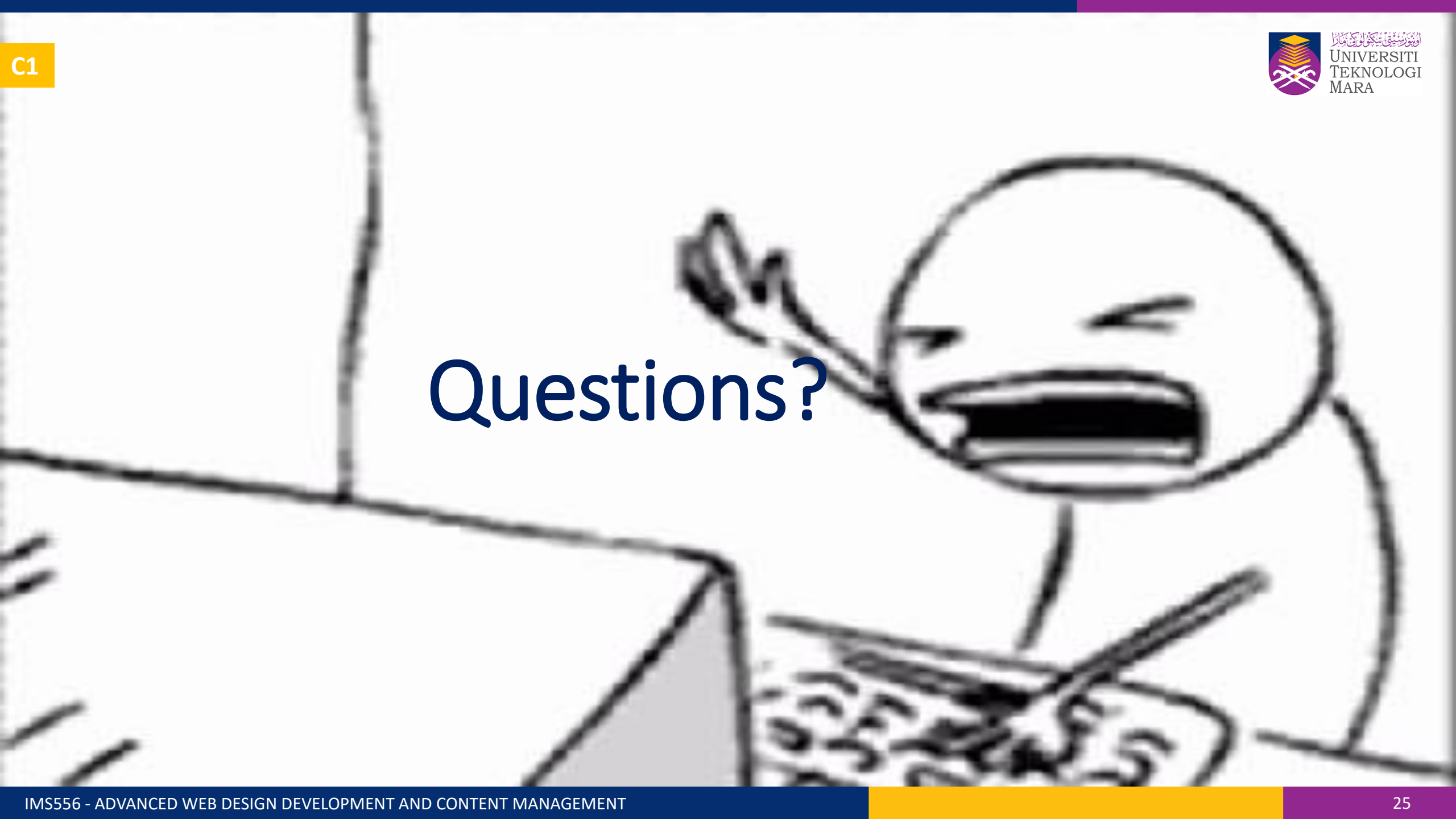
Conclusion

Web applications provide accessibility and convenience, allowing users to interact with services and information from any device with an internet connection. They support various functionalities, from simple tasks like simple browsing to complex operations like data analysis and project management. Using modern frameworks and tools in web application development ensures faster development cycles, improved efficiency, and easier maintenance.

A robust CMS supports collaboration, enhances productivity, and reduces costs by simplifying content updates and ensuring content consistency. Additionally, CMS tools can handle both structured and unstructured content, making them versatile solutions for various content management needs

Reflection

- What is your opinion on web application technology advancement?
- How AI influence the web application in future?
- How do web application development phases influence the quality and reliability of the developed web application?
- What are the disadvantages of using web applications?

A black and white cartoon illustration of a person with a large, round head and a wide-open mouth, appearing to shout or yell. The person is holding a laptop in front of them with both hands. The background is simple, with a few lines suggesting a room or desk.

Questions?